

Sampurna KC

Changunarayan, Bhaktapur | sampurnakc19@gmail.com | (+977) 9703548442 | github.com/TheZero19

Introduction

A computer engineer who is absolutely into video games, music, and chess.

Experience

Unity Game Developer, Avana Game Studio – Freelance July 2023 – May 2025

- Responsible for creating new game mechanics using Unity and writing API layer for using different services in the games.

Unity Game Developer, Yarsa Labs Pvt. Ltd. – Pulchowk, Lalitpur April 2024 – December 2024

- Responsible for studying and maintaining existing projects.

Technologies

Languages: C#, C/C++, Python, SQL

Game Engine: Unity Engine

Frameworks and Libraries: Django, Raylib, PyTorch, Plotly Dash, Pandas, Flower, Scikit-learn

Tools: Git, PlasticSCM, Firebase, JIRA, Figma, Bosca Ceoil, Krita

Education

Kathmandu University, BE in Computer Engineering 2019 – 2024

- GPA: 3.68/4.0
- **Electives:** Data Mining, Computer Vision

Capital College and Research Center, NEB 2017 - 2019

- GPA: 3.73/4.0

Golden Sungava English Secondary School, SEE Upto 2017

- GPA: 3.90/4.0

Major Projects

Dominoes, Yarsa Games

- Maintenance of Dominoes, the game with 100K+ userbase on Google Play Store.
- Worked with backend engineers to integrate their BaaS, Yarsa Play into the existing games.
- Changes related to game mechanics, UI and client-side service layer in Unity.
- Integrating and debugging the custom plugins related to Unity.
- Fixing build issues and errors for Android builds.
- Helping interns understand the fundamentals related to Unity and programming.

Avana Carrom, Avana Game Studio

- Physics and Game Mechanics of Carrom in Unity.
- Player profile system using Firebase Realtime Database
- Tournament Systems using Firestore and Remote Config
- Daily, All-Time and Tournament Leaderboards using Firestore
- UI design using Figma and its responsive implementation in Unity.

Other Projects

College Semester Projects

- Library Management System in Console (C++ and File handling)
- Music streaming platform (Python, Django, Html, CSS), *github – link – here*
- Medieval Adventure - Platformer and top down game (C#, Unity) *link – here*
- Sphere in Arms - FPP 1v1 game (C#, Unity, Photon), *link – here*
- Nalapani - TPP game (C#, Unity, Blender, Mixamo), *link – here*

College Mini Projects

- Pong Game - Computer Graphics Course (C++, Raylib), *github – link – here*
- Data Visualization - Data Mining Course (Python, Plotly, Dash, Scikit-learn), Hosted at *this – link*
- Posture Corrector - Computer Vision Course (Python, OpenPose), Detailed Paper at *this – link*

Mentorship

Federated Learning, Literature Recommendation System

- Provided remote guidance to an international university student studying in China, assisting with foundational concepts in Federated Learning for his thesis paper on Literature Recommendation System with Federated Learning.
- Helped him overcome his technical barrier related to programming by guiding him through the process of creating a recommendation system and its simulation using frameworks like PyTorch and Flower in Python.

Game Development and Programming Fundamentals

- Helped a student of Islington College in using the Unity Engine for his final year project related to a 2D platformer game.
- Mentored multiple people, supporting their learning related to Design Patterns, code optimization, and programming fundamentals.